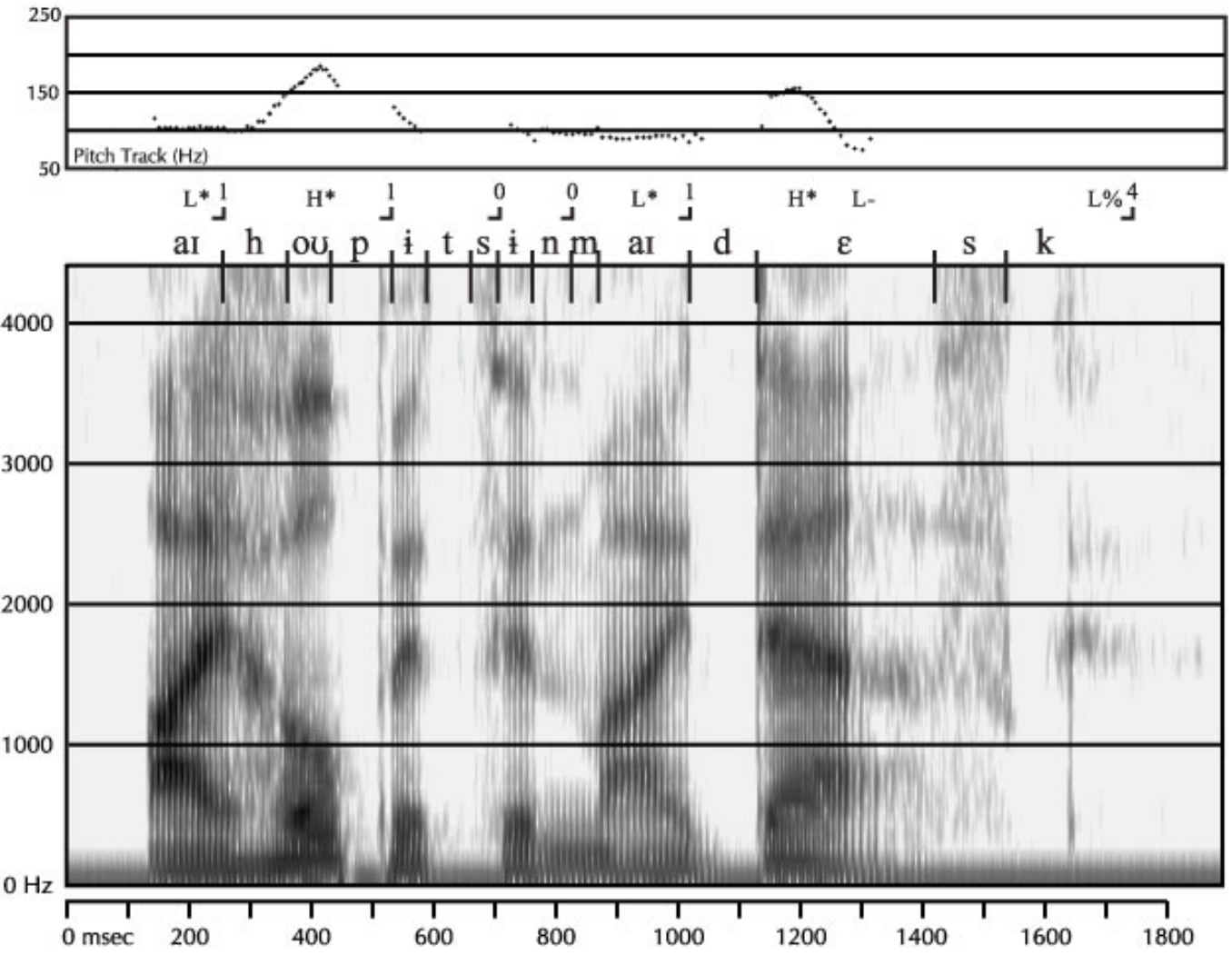
Development 2: Words, morphemes, constructions

Remember how messy speech is?



There are no physical boundaries between words.

Where there are pauses in the speech stream, these are the result of closures that are as likely to happen within as between words.

But we are really good at finding words in this “sea of sounds.”

How?

Harder question: How do we learn to do this?

Some cues to word boundaries

Words sometimes appear in isolation: “Hello.” “No.” “Please.” “Go.” “Stop.” etc., but not all words…

Phonotactics: Some sequences of phonemes are very unlikely to occur within words: “wordboundary,” “speechcode,” “*movement*.”

Stress patterns: In English, most words are stressed on the first syllable. (Jusczyk et al., 1999)

Kingdom passage

Your kingdom is in a faraway place. The prince used to sail to that kingdom when he came home from school. One day he saw a ghost in this old kingdom…

Device passage

Your device can do a lot. Her device only fixes things. My new red device makes ice cream. The pink device sews clothes….

7.5 month old Infants familiarized with “kingdom” showed a preference for the “kingdom” passage.

Infants familiarized with “device” did not show the same kind of preference.

In connected speech, stress may help signal the beginnings of words. (Of course this won’t work in every language, or for every word in English!)

Statistical learning of word segmentation from transitional probabilities

Another cue to word boundaries.

Consider: “Pretty baby.”

When you look at English words, the syllable /ti/ is very strongly predicted by the syllable /prI/, but this is less likely to be true for syllable pairs *across words:*

$$\frac{\text{(freq) } \textit{pretty}}{\text{(freq) } \textit{pre}} \quad >>> \quad \frac{\text{(freq) } \textit{tyba}}{\text{(freq) } \textit{ty}}$$

Statistical learning of word segmentation from transitional probabilities

So, a bit more abstractly, we can say that the likelihood of syllable B following syllable A is greater within than between words.

Of course this is probabilistic:

"What's that in the road a/head?"

"I scream you scream we all scream for ice cream."

Isolating statistical information from other cues (Saffran et al., 1996)

In order to show that transitional probabilities can be used to identify words, stimuli were tightly controlled:

- monotone (no changes in pitch, duration, amplitude, etc.)
- simple CV syllables (no phonotactic cues to word boundaries)
- presented continuously (no pauses)

A set of three-syllable “words” was created, (e.g., tokibu, gikoba, gopila)

A visualization…

tokibugikobagopilatipolutokibu
gopilatipolutokibugikobagopila
gikobatokibugopilatipolugikoba
tipolugikobatipolugopilatipolu
tokibugopilatipolutokibugopila
tipolutokibugopilagikobatipolu
tokibugopilagikobatipolugikoba
tipolugikobatipolutokibugikoba
gopilatipolugikobatokibugopila

A visualization…

tokibugikobagopilatipolutokibu
gopilatipolutokibugikobagopila
gikobatokibugopilatipolugikoba
tipolugikobatipolugopilatipolu
tokibugopilatipolutokibugopila
tipolutokibugopilagikobatipolu
tokibugopilagikobatipolugikoba
tipolugikobatipolutokibugikoba
gopilatipolugikobatokibugopila

Testing for an effect of transitional probabilities

After listening to a long stream of syllables, it's trivial to show that infants are more familiar with sequences they have heard than ones they haven't.

Two additional details are important:

1. Half of the words were repeated more frequently than the other half.
2. Test for familiarity was against "part words" that were matched for frequency of occurrence with real words.

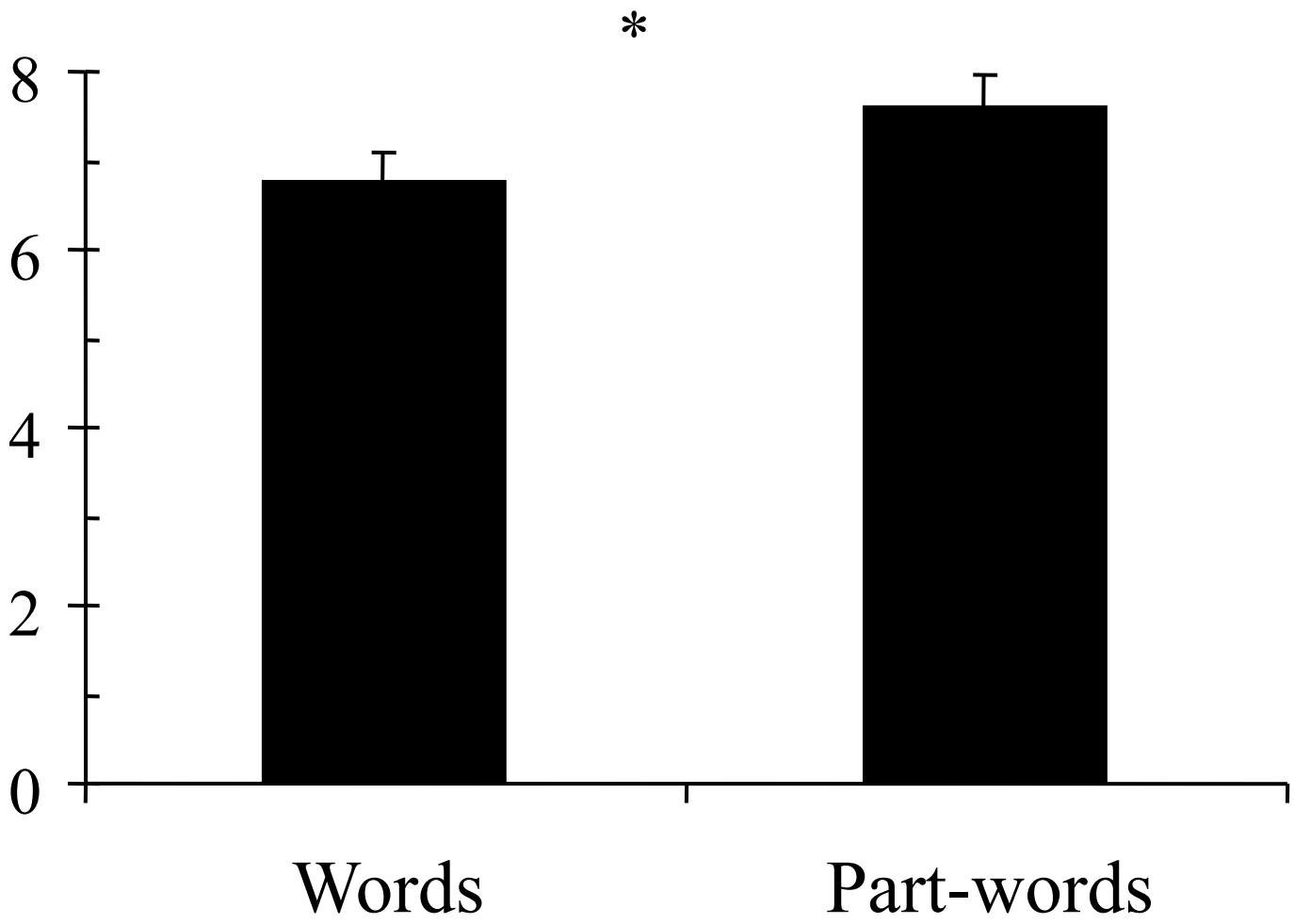
Example:
toki**bu giko**ba **gopila** gikoba tokibu **gopila** toki**bu giko**ba gikoba tokibu

Here GOPILA and BU-GIKO both appear twice, but the probability of GI given BU is 2/4, whereas the TPs for GOPILA are both 2/2.

A novelty effect

Infants looked longer in the direction of “part word” stimuli.

This suggests that they had learned the statistical regularities of the speech stream, and were not merely picking up on simple frequency of occurrence.

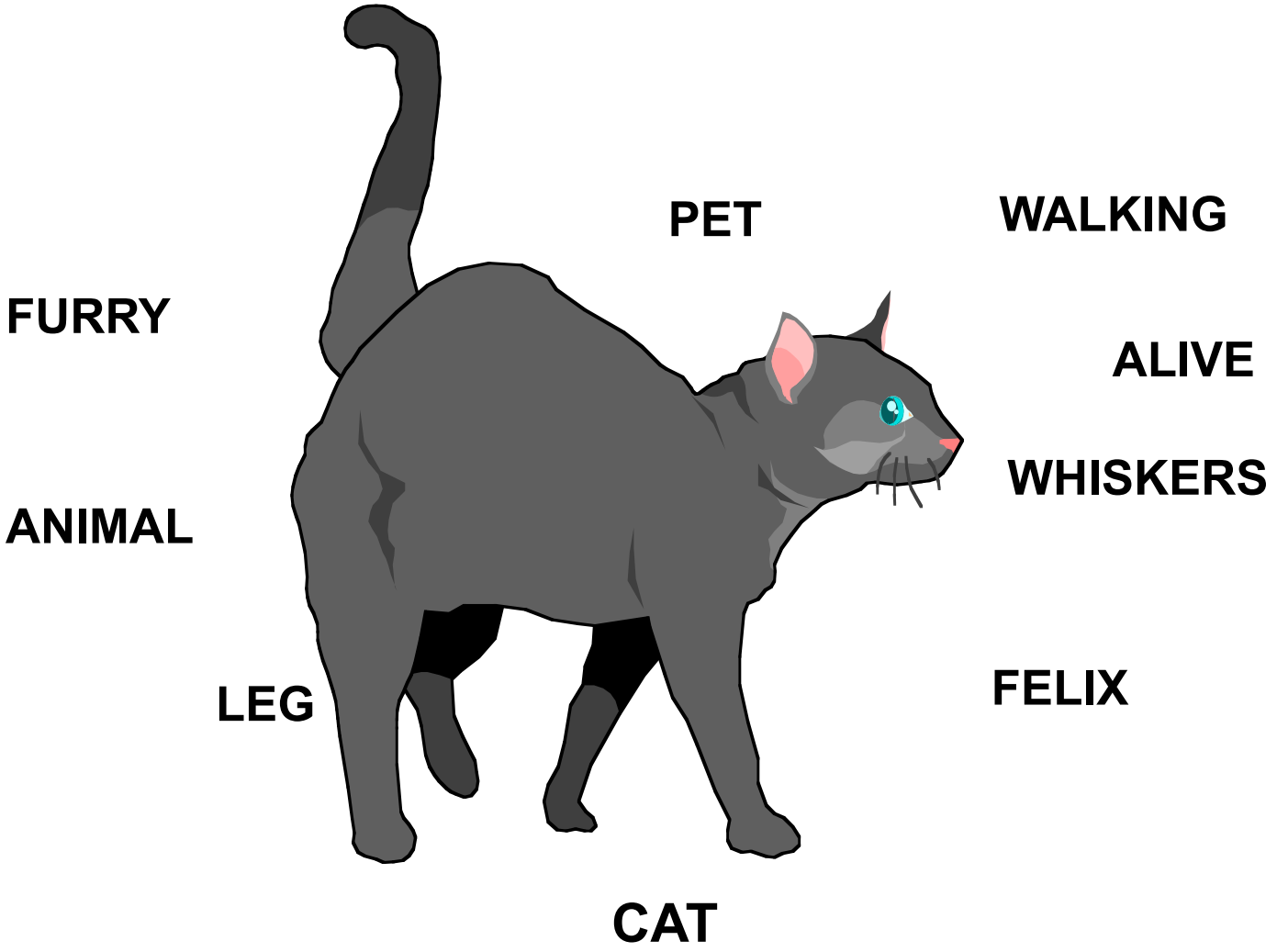


Identifying Word Referents
(or:…now that kids can “find” the words, how do they learn what they mean?)



CAT

The inductive problem of early word learning



See,e.g, Quine's formulation of the "gavagai" problem:
https://en.wikipedia.org/wiki/Inscrutability_of_reference#Illustration_by_the_use_of_gavagai

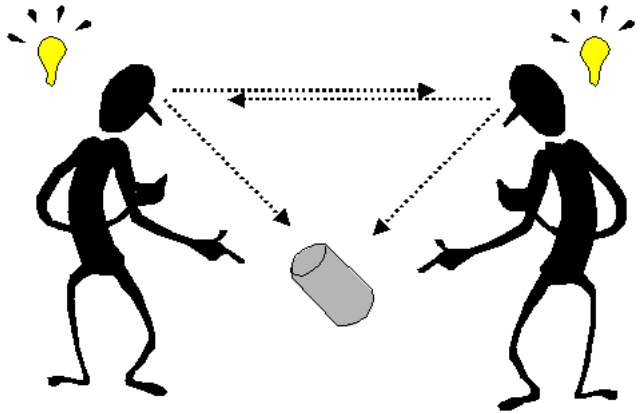
How do children acquire word meanings?

Joint attention:

Sharing an attentional focus with another

Gaze following emerges around 9 months of age

Signals an understanding of reference



How do children acquire word meanings?

Joint attention:

- Optimizes word learning
- Baldwin et al.
  - *Joint attention group*: adult interacting with baby labeled the object the baby was playing with
  - *No joint attention group*: voice from behind a screen labeled the object the baby was playing with

Results:

- 18-month-olds learned the novel word in the *Joint attention* group but not in the *No joint attention* group.

How do children acquire word meanings?

How do children narrow the hypothesis space for new words?

Intentional cues from the speaker

e.g., eye gaze direction



Baldwin: 16-17 and 18-19 month olds taught new words

Follow-in labeling: looked at toy that infant was focused on

Discrepant labeling: looked at toy in a bucket

Gaze Direction



How do children acquire word meanings?

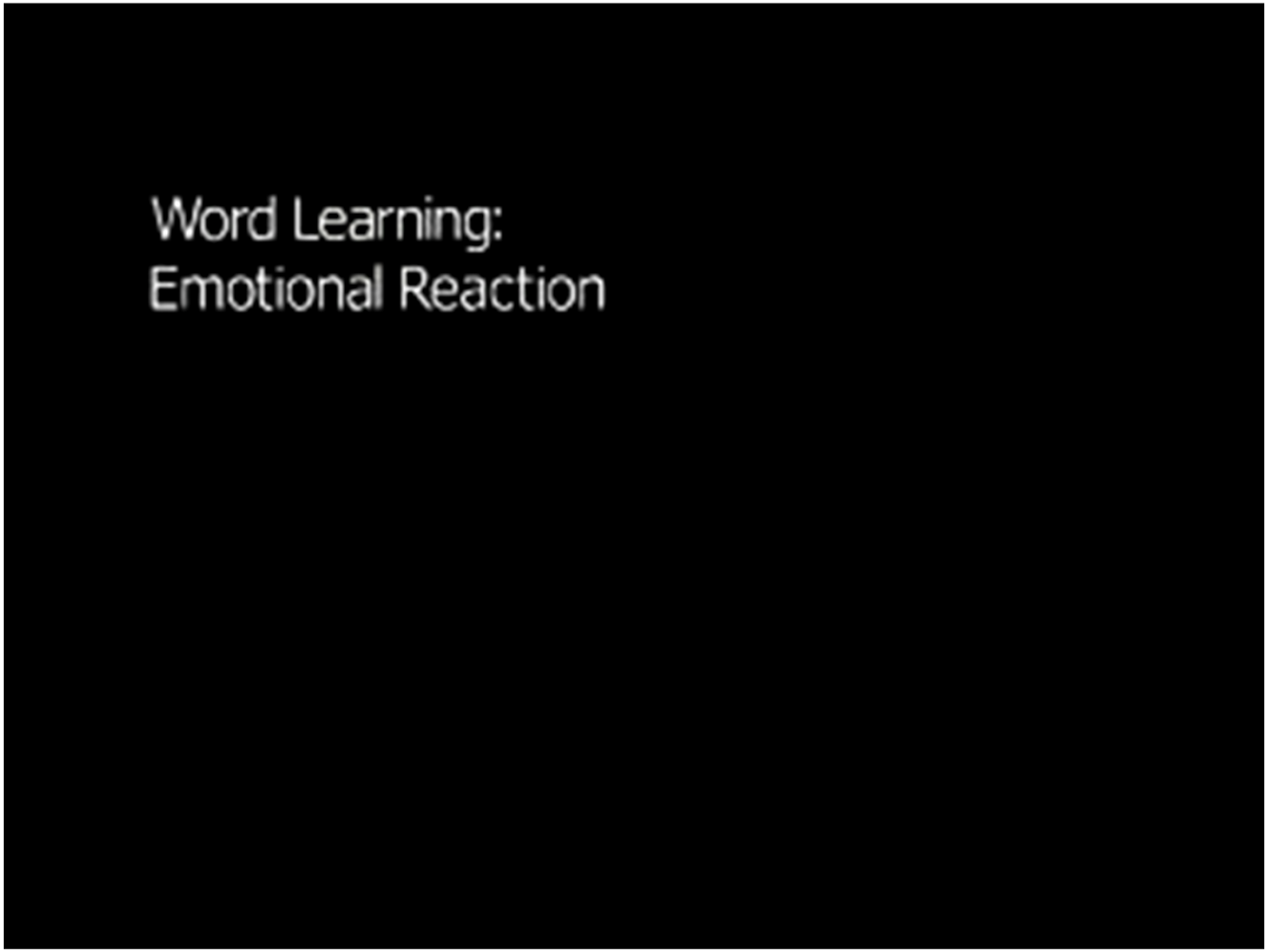
Intentional cues from the speaker

e.g., affective and/or behavioral cues

Akhtar et al.

Infants used adults' facial expression to identify the intended referent of a novel word.

Intentional Cues



How do children acquire word meanings?

Word learning strategies:

Default strategies that children bring to the word learning situation

Function to narrow the hypothesis space for new words

- e.g., the whole object assumption

How do children acquire word meanings?

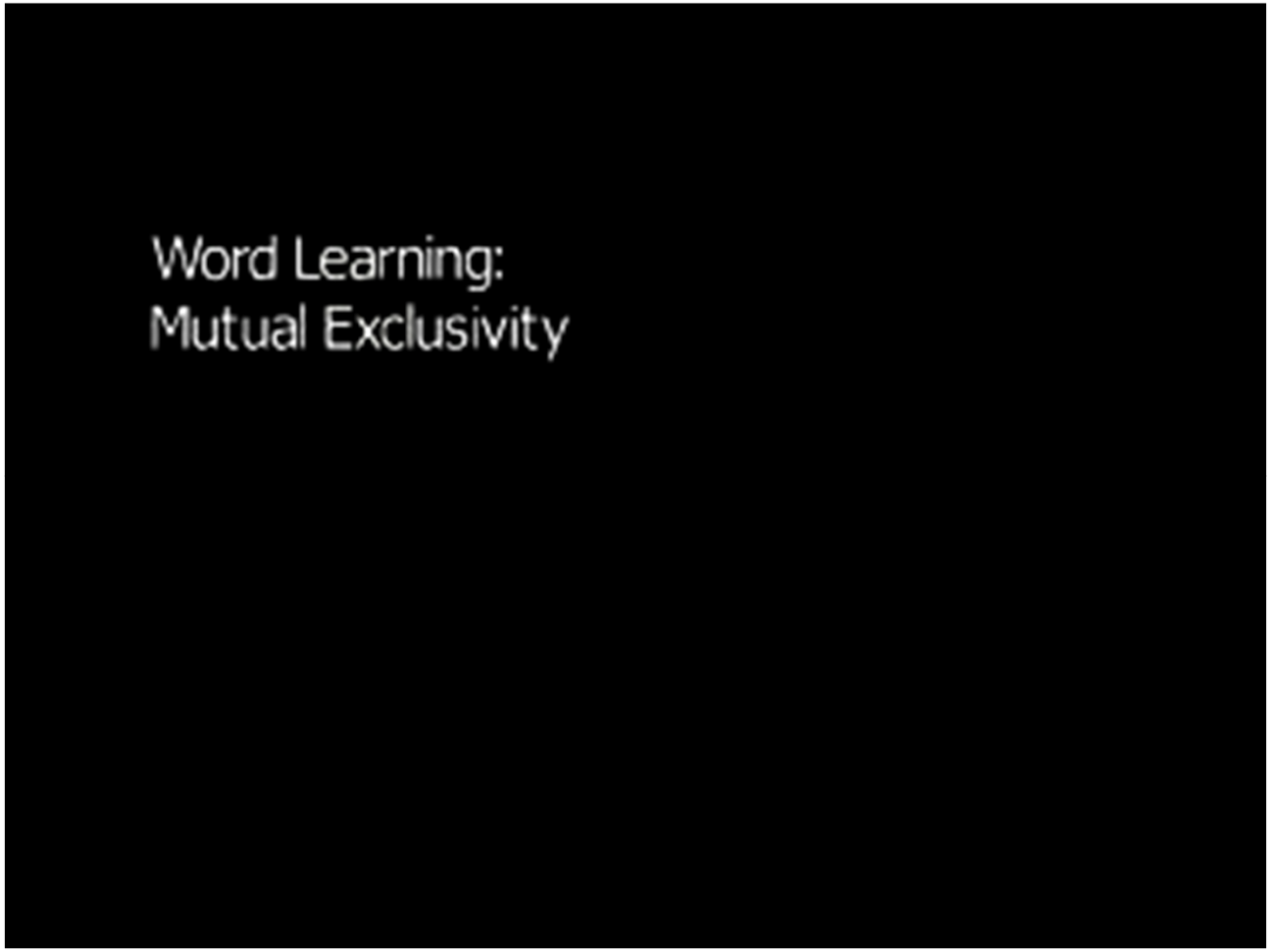
e.g., Mutual Exclusivity:(Markman, 1989)

- as a default, objects can have only one name
- Implication: new words map to nameless objects
- Emerges during the second half of the second year





Mutual Exclusivity



How do children start putting words together?

In production, it can be difficult to distinguish what's going on when children produce only one, two or three words at a time.

Menn distinguishes among:

- "just a bunch of relevant words"
- formulas
- word combinations
- stable constructions (not necessarily adult-like)

In practice, a full description of the context in which an utterance is produced is necessary (but often not sufficient!) to distinguish among these.

Up!

Pick you up!

Mommy up!



Comprehension vs. production

Even children at the *one-word stage* in production are able to use basic transitive word order in *comprehension*

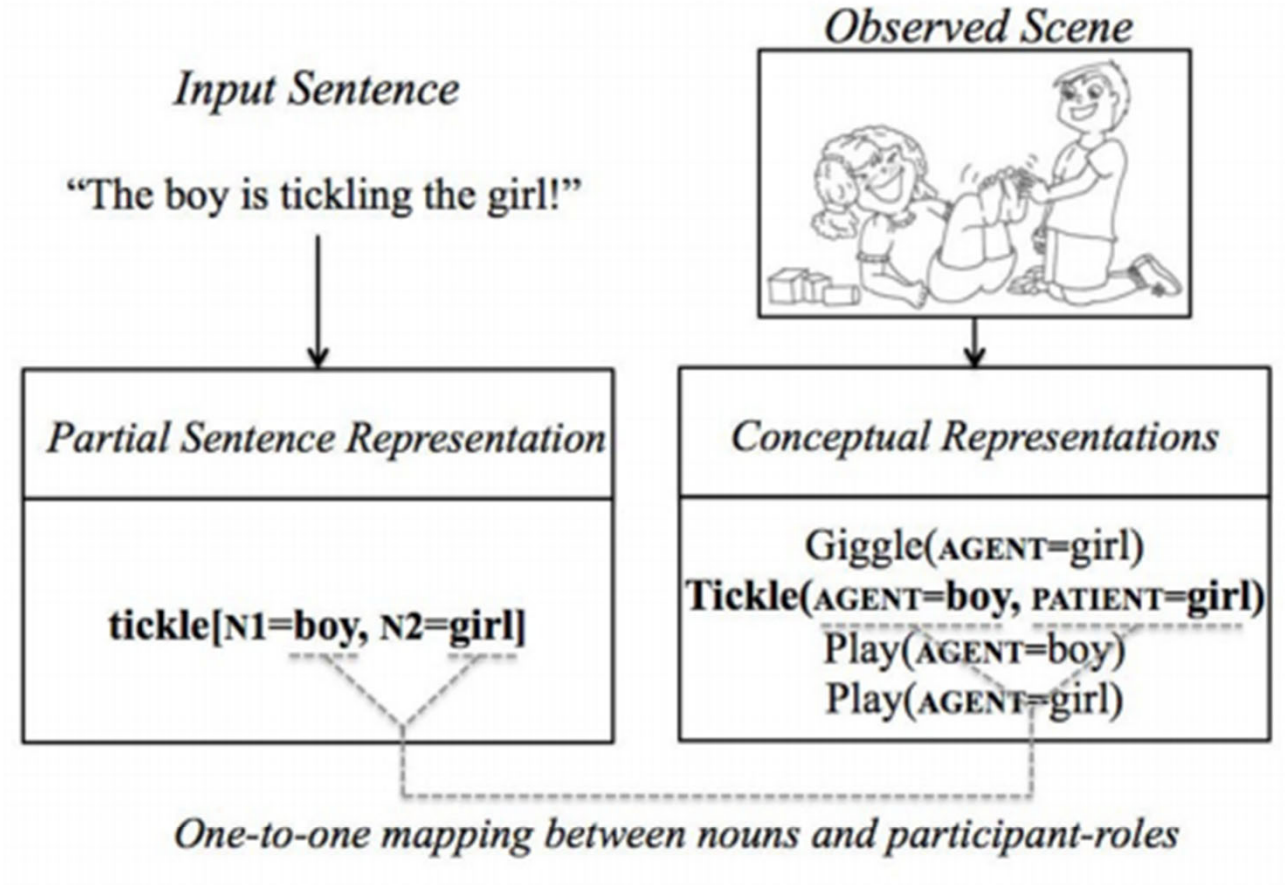
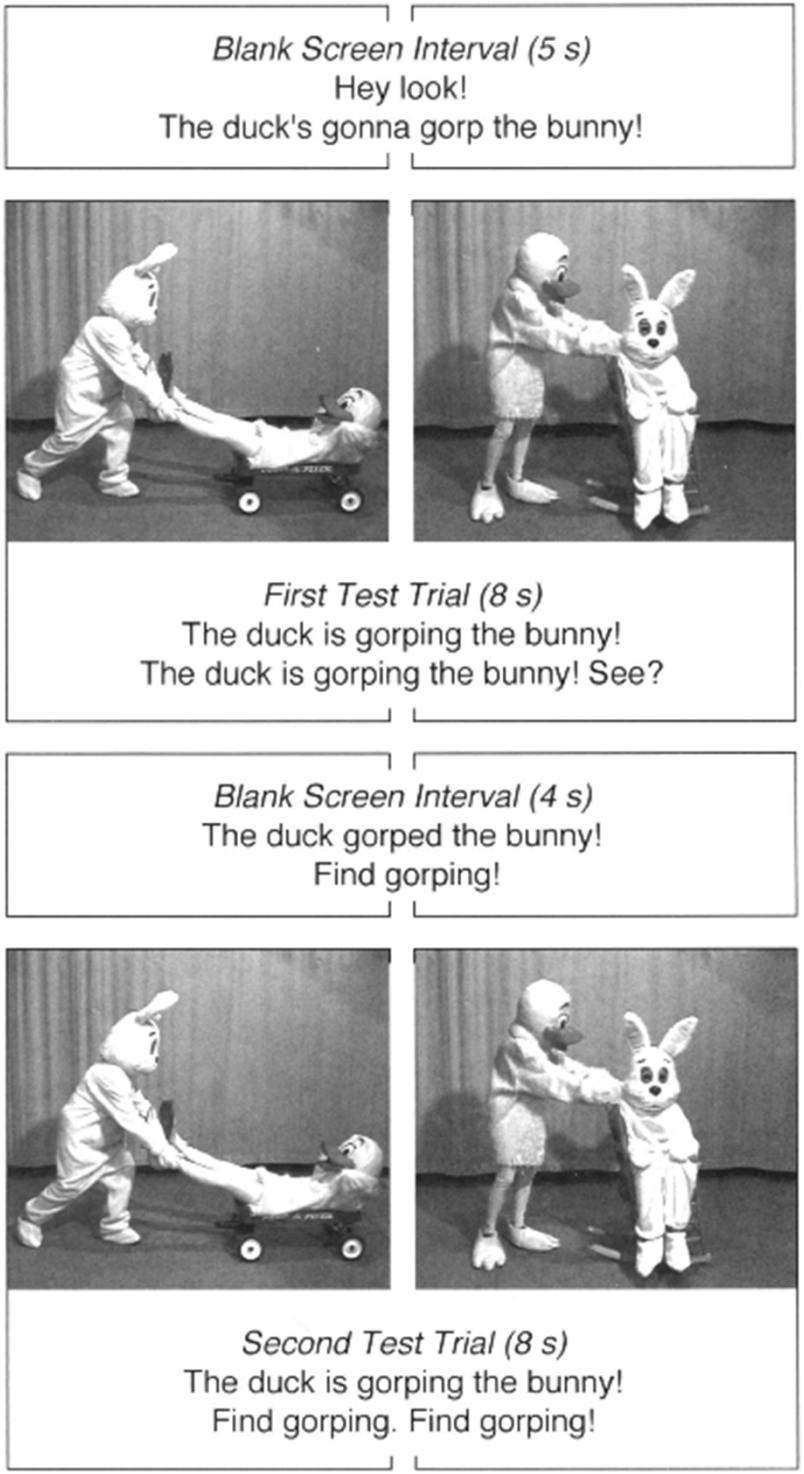


Figure 1.
Schematic diagram of early syntactic bootstrapping via the structure-mapping account.



A **generalization** test:

If children have really learned the transitive structure, they will generalize it to new verbs.

Think about what you need to know in order to select the picture that could depict “The duck is gorping the bunny” from these two novel actions.

Gertner et al (2006) showed that two-year olds can understand this: They look significantly longer at the duck acting on the rabbit than vice-versa.)

At this age, kids typically have a mean length of utterance (MLU) of around 2.

Their three morpheme utterances might be like “Duck gorp bunny.”